

ANIMAL GENEALOGY

In the 1950s, a revolutionary new way of classifying organisms was proposed. Called phylogenetics, it allowed taxonomists to investigate evolutionary relationships between species by placing them in hierarchical groups called clades.

Phylogenetics, also known as cladistics, is based on the work of entomologist Willi Hennig (1913–1976). He assumed that organisms with the same morphological characters must be more closely related to each other than to those that lacked them. Therefore, these organisms must also share the same evolutionary history and have a more recent ancestor in common. Like the traditional taxonomy of Linnaeus, this method of classifying organisms is hierarchical, but due to the volume of data involved, computers are used to generate the family trees, known as cladograms.

For a morphological feature to be useful in cladistic analysis, it must have altered in some way from the so-called “primitive” ancestral condition to a “derived” one. For example, the legs and paws of most carnivores are considered primitive in the cladogram on the opposite page, compared to the derived condition of flippers in the seals, fur seals and sea lions, and walruses. This derived character, called a synapomorphy, is useful because it is shared between at least two taxonomic groups, suggesting

they are more closely related to each other than to groups without flippers. Characters that are unique to one group are useful for recognition purposes but say nothing about relationships. So cladistic analysis is based entirely on the identification of synapomorphic characters.

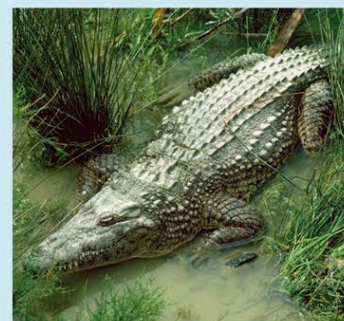
UNDERSTANDING ANCESTRY

The more derived characters organisms have in common, the closer their relationship is assumed to be. For instance, brothers and sisters look more alike than other children – they may have the same eyes, the same chin, and so on. This is because in terms of common ancestry they have the same parents but are only distantly related to other people.

Cladistics is now most commonly based on genetics – except in the study of fossils – and has exposed some unexpected shared ancestry. For example, a genetic cladogram surprisingly revealed that the whale’s closest living relative is the land-dwelling hippopotamus – a relationship that Linnaeus would certainly not have expected.

SCIENCE OUTGROUPS

The first step in doing cladistic analysis of a group is to choose a closely related but more primitive species (the outgroup) for comparison. This allows derived characters to be distinguished from primitive ones. For example, to work out a phylogenetic tree of birds, a crocodile might be chosen as the outgroup, as both birds and crocodiles belong to the clade Archosauria.



A PRIMITIVE BIRD RELATIVE

✓ CLOSE RELATIONS

The giraffes and kudus are both even-toed ungulates, so are more closely related to each other than to the odd-toed zebras. As fur-bearing mammals, all three are more closely related to each other than to the feathered birds around them.

